Fiscal Year:	FY 2012	**Task Last Updated:**	FY 01/15/2014
PI Name:	McKay, David Ph.D.		
Project Title:	Geology, Geochemistry and Lithology Science Support Activities		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Environmental health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	david.s.mckay@nasa.gov, erin.s.connell@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-483-5048
Organization Name:	NASA Johnson Space Center		
PI Address 1:	2101 NASA Parkway, KA		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:	PI is deceased (Ed., Feb. 2013)		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/02/2006	**End Date:**	09/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Sullivan, Thomas	**Contact Phone:**	
Contact Email:	thomas.a.sullivan@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Changed end date to 9/30/2012 since this portion of project has been completed (Ed. 1/15/14) NOTE: End date changed to 9/30/2013 per John James (Ed., 8/24/2012) NOTE: End date changed to 9/30/2011 per JSC HRP info--MTL dtd 10/7/11 (Ed., 10/28/11) NOTE: Start/end dates changed to 10/2/2006-12/31/2010 (previously 4/30/2006-1/31/2011) per B. Woolford/JSC via S. Steinberg-Wright/JSC (9/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Taylor, Lawrence (University of Tennessee –Knoxville)		
Grant/Contract No.:	Directed Research		
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Performance Goal Text:			

Task Description:	The Lunar Airborne Dust Toxicity Assessment Group (LADTAG) has been assigned the task of evaluating dust hazards to humans, and recommending exposure standards for lunar dust suspended in breathing air. Over the 40 years that have transpired since the last Apollo crew left the lunar surface, much has been learned about the toxicity of small particles. Based on experience with industrial minerals such as sandblasting quartz, we know that even if a dust is not hazardous due to a three-day exposure, it may nonetheless be quite toxic when the exposure covers a longer duration. The Lunar Airborne Dust Toxicity Assessment Group (LADTAG) was chartered by NASA to set an exposure standard for 6 months of episodic exposure to lunar dust. In order to accomplish this, it is necessary to learn how lunar dust toxicity compares to both toxic and non-toxic terrestrial dusts. Dust exposure standards, similar to those established for quartz and asbestos on Earth, will determine the design, mass, power, and cost of airlocks, ports, and dust control systems that are incorporated into lunar habitats and shirtsleeve rovers. The Lunar Airborne Dust Toxicity Advisory Group (LADTAG) Geology team, led by JSC senior scientist David S. McKay, encompasses expertise in geology, geochemistry, and geophysics, with a combined total of over 100 years of lunar research. This team provides three major research thrusts for the LADTAG research effort: (1) Evaluation and Recommendations; (2) Isolating the respirable fraction of lunar dust for inhalation toxicology studies, and (3) Lunar soil simulant development. (1) Evaluation and Recommendations: Lunar dust in situ is subjected to processes that likely cause grain surfaces to become chemically reactive, potentially increasing the toxicity of the dust. Moreover, lunar dust contains metallic iron in spheroids of 30 nanometers and smaller diameter. The effect of this nanophase iron on human tissue and cells is yet to be determined; such material does not occur on Earth and it is thus unknown whether humans have any specific evolutionary ability to deal with it. The detailed characteristics of sub-micrometer lunar dust are only poorly known, and this is the size range of particles that are of greatest concern for our toxicology studies. In addition to being potentially hazardous to humans, lunar dust is detrimental to equipment. Electrostatic charging of the dust due to UV irradiation and solar-wind-proton implantation must be characterized in order to design methods of mitigating these effects. Our team will provide high-fidelity simulants by continuing our research in the effects of radiation, proton bombardment, and triboelectric charging. One of the most unusual features of actual lunar dust is the presence of minute iron spheres, up to 30 nanometers in diameter, which often cover the surface of the grain. Our team has simulated these iron spheres in a silicate matrix for use in toxicology studies. Our activities in support of Evaluations and Recommendations include basic research on chemical activation, passivation, size and shape analysis, and simulating the unique qualities of the surfaces of lunar dust grains as it will be found on the lunar surface. In addition to providing knowledge about past research on lunar dust, our team will be conducting new experiments to address unknowns that have not previously been considered. For example, we plan dissolution studies, both with dust simulant and with actual lunar dust, to learn more about how dust particles behave when ingested. This provides critical background information for toxicology studies. (2) Isolating the respirable fraction of lunar dust for inhalation toxicology studies: Inhalation toxicology studies using lunar dust will be used to determine the allowable exposure limits. These experiments must be performed in a standardized manner so that the results obtained can be realistically compared to the results of experiments using other substances (such as silica dust). Consequently it is necessary to separate the respirable particles from a lunar sample, because the typical laboratory equipment for inhalation studies depends on having particles in the proper size range at the start of the experiment. Our efforts focus on dust in the respirable range and include the particle size distribution, mineralogical and morphological habits of the dust in lunar-habitat conditions, dust reactivity and passivation in applicable conditions, and direct cell interactions with nano-scale-iron-rich minerals and glasses. This allows LADTAG to define risks and establish appropriate standards for manned lunar missions. These analyses allow us to prepare lunar soils with the targeted size distributions and properties for use by the toxicology groups for inhalation and cell reaction studies. An additional by-product is the development of improved techniques for separating and handling nano-scale particles for Scanning Eletron Microscope (SEM), Transmission Electron Microscope (TEM), and elemental analysis. These requirements have inspired us to develop a method of separating the respirable fraction of lunar dust from a bulk soil sample, without using sieves, which allows maximum recovery of the available dust in the sample. The total amount of lunar sample available for this study is less than 300 grams. The developed method uses pneumatic transport with ultrapure nitrogen as the carrier gas; thus the lunar material is not exposed to water or other liquids. Our separation method incorporates a small fluidized bed as well as gas impactor size separation steps and a cyclonic separator. Analysis of the separated material with a Microtrac™ laser diffraction system has been used to validate the process. (3) Lunar soil simulant development: The geology team is developing new methods to simulate the chemically active surface of dust grains. Specifically, the areas of concern for LADTAG are the physical and chemical properties of the finest fraction (less than 3 μm) of lunar soil (“dust”), and the interaction of lunar dust with biological as well as mechanical systems. Before the hazards of lunar dust can be managed, they must be understood. The path to understanding the hazards includes both experimental and analytical work. The experimental work cannot be performed on actual lunar samples without first validating the proposed techniques using an appropriate simulant material. By developing an appropriate dust simulant, the Geology Team provides the basis for all other areas of research within LADTAG.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal.
Research Impact/Earth Benefits:	The separation method can be used for other purposes, and can be modified to handle larger or smaller quantities of material. It is particularly useful for isolating statistically representative particles for Transmission Electron Microscopy (TEM) studies. The system could also be modified to collect a different size fraction. Follow-on applications for this system include actual lunar surface use, perhaps as part of a pneumatic mining and soil transport system on the moon as the central core for lunar ISRU hydrogen extraction, and industrial waste clean-up (Cuero and McKay 2007).

Task Progress:	ED. NOTE 1/15/2014: Changed end date to 9/30/2012 since this portion of project has been completed and PI is deceased. Final progress information from Human Research Roadmap (http://humanresearchroadmap.nasa.gov/) Aims: Secure from the Curator and Planning Team for Extraterrestrial Materials (CAPTEM) samples of lunar dust. Develop a dry method for separating respirable size particles from the samples of authentic native dust. Develop a protocol for producing respirable size particles by grinding courser lunar dust. Determine mass median aerodynamic diameters and mineralogy of the respirable lunar dust samples. All aims have been realized. Bulk samples of lunar dust were obtained from CAPTEM and respirable size particles of native lunar dust were separated using a new pneumatic technique developed by the Geology Team. This technique was used to separate additional respirable particles from coarse lunar dust, which had been reduced by grinding. Mass median aerodynamic diameter of separated particles were measured with particle sizers and confirmed to be in the respirable range. Respirable particles were utilized in studies with rodents that assessed the toxicity of the lunar dusts. The studies addressed the gap “What is the toxicity of lunar dust in respiratory system". Integration/Unique Aspects: The products of this task will be utilized in other tasks that address Gaps AEH2, AEH4, and AEH5.
Bibliography Type:	Description: (Last Updated: 06/08/2010)